From business question to application

Homework: steps 1 to 3 from lecture 2 *'Asking the right questions'*

Group	R3			
Dataset / case	ReE: Research CRM Network Graph			
Members	1. Siemen	2. Mikho	3. Patrick Ngibuini	

Before we start building the dashboard, it is important that we know what to build. Fill In the following template: one per group. Take enough time to ensure you understand which questions to ask: you can make the world's most fancy dashboard, if it answers the wrong question, nobody will actually use it.

Step 1 · Understand the business question

Goal: one paragraph with the business problem, the decision it supports and who decides.

Business questions

Ask more than "what do you want to see?". Write your own open-ended question per topic, then fill in what the business owner would answer. Not all of these topics will be equally important. If you really can't think of a good question, then maybe that is a sign that this particular topic is not very relevant for your case. On the other hand, if you have many questions for a particular topic, you know to pay more attention to it. Writing 'not relevant' for a topic is also a good answer, as is writing multiple questions.

Topic	Your question	Your answer
Problem *What is going wrong or unknown? Why now? What does the business owner want to solve?*	What is the biggest thing you don't know about your own department today? What happens when a project ends?	We don't have one place that shows which partners we have, who in our team knows them and how strong that link is. When a project ends, the contact often disappears with it, and three years later nobody remembers we already worked with that company.
Decision *What will someone do differently after using the dashboard?*	Which concrete choices should the app help you make?	Who to contact at a partner organisation and through which colleague, which partners we should keep warm before a project ends, and which people will be free to start on a new proposal. In short: choosing people and partners for the next project.
People *Who will use it? Who else sees the result?*	Who will open the app, and who receives what comes out of it?	The business developer (valorisation) who looks after partnerships, the financial and project administrator who keeps projects and budgets up to date, and new researchers who need to find their way. The head of department gets the overviews.
Workflow *How do you handle this today? Where do you get stuck? What could be better?*	Walk us through the last time you looked for a partner or an expert. Where did it go slow?	We searched old mails and Excel lists and asked the group leads. The slow part is checking whether the information is still correct: is this project still running, does this contact still work there, is that colleague still here?

Rhythm *Daily, weekly, once per project? How often will people look at the dashboard?*	How often would each person use it?	The business developer weekly, the administrator whenever something changes (a new project, a status change, someone leaving), and researchers mostly in the weeks before a call deadline. New researchers a few times in their first month.
Exploration *What do you compare, filter, zoom into? Which "what if" do you want to answer?*	Which comparisons and filters do you need? Which "what if" questions come up?	Compare research groups and funding programmes, filter on status, period, company type and contact role, and click a company or contact to see all its links. What if: if this project ends, which partners do we still have a running link with? And how many people become available in the coming months?
Data *What data exists, at what detail, updated how often?*	Which data do you have, how detailed, and is it up to date?	seed.json and the CSVs: 10 projects, 20 employees, 15 contacts at 6 companies (8 departments), 16 skills and about 150 relations with budget, role, percentage and level. When we looked at the data we found that the status is not always up to date compared to today: CITYPULSE, FRESHCAST, SENSEHOME and WARMWIJK are still "running" while their end date has passed, SENIORLINK is still "approved" while its start date is in the past, and AIAUDIT is "submitted" with a start date that already passed. Some contacts are in several projects (Anke Janssens of Stad Kesselberg in three). Four skills (explainable AI, NLP, data engineering, dashboarding) belong to only one expert each.
Success *How will you know the dashboard helped?*	When would you say the app was worth it?	When the business developer can show all links with one partner in under a minute, when the status mismatches like the ones above are caught by the app instead of by accident, and when no partner relationship is lost because a project ended without anyone noticing.

Summary

➔ What will be the main focus/problem that this dashboard answers?

[Client] needs to *[decision/action]* because *[problem]*. *[Users]* will use the app *[how often]* to *[main tasks]*, based on *[data]*.

The UCLL research department needs to keep its partner relationships alive and choose the right people and partners for new projects, because today that knowledge is scattered and out of date, so links with companies get lost when projects end. The business developer (weekly), the project administrator (whenever data changes) and researchers (around call deadlines) will use the app to explore who is connected to which partner, check which projects and links are still active, detect gaps such as expertise held by one person, and plan ahead for projects that are ending, based on the projects, people, skills, groups, companies and contacts in seed.json.

Tip: if your summary only contains numbers to monitor ("see total sales"), you have a KPI dashboard. Find tasks: comparing, explaining, locating, planning, simulating.

Step 2 · Identify users and their tasks

Goal: 2 to 3 user profiles with 2 to 3 tasks each.

Users

User (role)	Expertise	Context (where, when, how often)
Business developer (valorisation, partnerships)	Knows the partners and the market well, knows the data less. Thinks in companies and contacts, not in tables.	Weekly at the office, and before meetings with a partner or a funder.
Financial and project administrator	Knows every project, its budget and dates in detail. Comfortable with data entry and checks.	At the desk whenever a project is approved, starts, ends or changes budget. A few times per month.
New researcher	Knows their own field but not the department, its people or its partners yet.	In the first weeks after joining, and later when looking for a colleague to work with on a proposal.

User stories

Useful verbs: locate, compare, filter, drill down, explain, detect, simulate, export.
A story with only "see" or "view" is too vague. Ask: see it in order to do what?

#	User	As a [role], I want to [action in the app] so that I can [decision or outcome].
S1	Business developer	As a business developer, I want to locate the contacts who are involved in several of our projects, so that I can treat them as key relations and keep them informed.
S2	Business developer	As a business developer, I want to drill down into one company and see its departments, contacts, projects and our research groups, so that I can prepare a meeting with that partner.
S3	Business developer	As a business developer, I want to simulate a project ending and see which partners no longer have a running project with us, so that I can propose a follow-up in time.
S4	Administrator	As a project administrator, I want to detect projects whose status does not match their dates, so that I can correct them and keep the data trustworthy.
S5	Administrator	As a project administrator, I want to compare the budget per research group and funding programme, so that I can report to the head of department.
S6	Administrator	As a project administrator, I want to filter projects on group and period on a timeline, so that I can see which projects end soon and plan the reporting.
S7	New researcher	As a new researcher, I want to filter colleagues on a skill category and level, so that I can find who to work with or learn from.
S8	New researcher	As a new researcher, I want to detect skills that only one colleague has, so that I can see where my own expertise adds something to the department.
S9		Not used.

Step 3 · Derive the user questions

Goal: 4 to 8 numbered user questions. Each one is asked while working in the app and becomes at least one feature later.

Per user story: extract, break down (where? what does it look like? why? what if I act?), refine, review. If a question does not help the decision from Step 1, drop it.

> **Make the variable part explicit**
> **KPI:** "What was the booking rate in December?"
> **App:** "What was the booking rate for [selected customer group] during [selected period], compared to [baseline period]?"

Type: descriptive = what happened, where? · diagnostic = why? · predictive = what will happen? · what-if = what if I change X?

Acceptance criterion: one per question. Example: given a station is selected, when the user changes the date range, then the daily profile (Q2) updates.

#	User question, variable parts in [brackets]	Type	Acceptance criterion
Q1	Which external contacts take part in at least [N] projects with status [status], in which roles, during [date range]?	☒ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if Story: S1	**Given** the seed data is loaded and rejected projects are hidden **When** the user sets N to 3 in the key contacts view **Then** Anke Janssens (Stad Kesselberg) appears with CITYPULSE, DOCMINE and SENIORLINK and her role in each, and she is highlighted in the network
Q2	Through which departments, contacts, projects and research groups are we connected to [selected company]?	☐ Descriptive ☒ Diagnostic ☐ Predictive ☐ What-if Story: S2	**Given** a company node is visible **When** the user double clicks Stad Kesselberg **Then** the ego view shows its two departments (Smart City Office, Welfare Services), their contacts, the three projects (CITYPULSE, DOCMINE, SENIORLINK) and the groups DSAI and HCD, and the detail panel lists them
Q3	Which projects have a status that does not fit their dates on [reference date]?	☐ Descriptive ☒ Diagnostic ☐ Predictive ☐ What-if Story: S4	**Given** the reference date is set to today **When** the user opens the data check tab **Then** the table lists CITYPULSE, FRESHCAST, SENSEHOME and WARMWIJK (running but end date passed), SENIORLINK and AIAUDIT (start date passed), with a link to edit each project
Q4	Which projects of [research group] run during [period], and when does each one end?	☒ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if Story: S6	**Given** the timeline view is open **When** the user selects research group SET and the period 2023 to 2026 **Then** the timeline shows WARMWIJK and SENSEHOME as bars from start to end date, and ENERGYCOMM is hidden because it ended in 2022

Q5	Which skills in [skill category] are held by at most [N] employees at level [minimum level]?	☒ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if Story: S8	**Given** the seed data is loaded **When** the user selects category ML, N = 1 and level expert **Then** the chart shows Explainable AI (Bram Wouters) and Natural language processing (Els Vermeulen) as single expert skills
Q6	How is the total budget of [funding programme] split over the [research groups], for projects with status [status]?	☒ Descriptive ☐ Diagnostic ☐ Predictive ☐ What-if Story: S5	**Given** the overview tab is open **When** the user selects VLAIO TETRA and excludes rejected projects **Then** a sorted bar chart shows the budget per group as the sum of the funds_group links, and SHELFVISION is no longer included
Q7	If [selected project] ends on [end date], which companies and contacts have no other running project with us?	☐ Descriptive ☐ Diagnostic ☐ Predictive ☒ What-if Story: S3	**Given** the user selects XAICARE **When** the user moves the end date slider and clicks "Simulate" **Then** the before and after network shows that Zorgnet Hageland has no other project left, while VIDT only keeps a link through the submitted AIAUDIT proposal
Q8	How many employees of [research group] will have no running project in the next [N] months, based on the planned end dates?	☐ Descriptive ☐ Diagnostic ☒ Predictive ☐ What-if Story: S6	**Given** the timeline view is open and group SET is selected **When** the user sets N to 6 months **Then** the app shows the number and names of employees whose projects all end in that window, as a forecast based on the dates, not on new projects